PRODUCT USER MANUAL

For WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002

Issue: 1.0

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	2025/05/15	All	New format of the PUM including the information of the last version of the CMEMS-WAV-PUM-014-001-002-003-004-009-V3. EIS November 2025: modification of dataset name to be compliant with MOi rules (addition of '-i')	M. De Carlo	P. Zunino

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GLOSSARY AND ABBREVIATIONS

ADD	Architecture and Design Document
AL	Saral/AltiKa
C2	Cryosat-2
CalVal	Calibration/Validation
CF	Climate Forecast (convention for NetCDF)
CFO	CFOSAT
CLS	Collecte Localisation Satellite
CMEMS	Copernicus Marine Environment Monitoring Service
DT	Delayed Time
DU	Dissemination Unit
GDR	Geophysical Data Record
GOP	Geophysical Ocean Product
HY-2B	Hai Yang 2B
HY-2C	Hai Yang 2C
IGDR	Intermediate Geophysical Data Record
J3	Jason-3
KPI	Key Production Indicator
L2/L2P/L3/L4	Level 2 / Level 2 Plus / Level 3 / Level 4
LRM	Low Resolution Mode
NetCDF	Network Common Data Form
NOP	NRT Ocean Product
NRT	Near Real Time
OE	Orbit Error
OGDR	Operational Geophysical data record
OLA	Operational Level Agreement
PIT	Product Information Table
PLRM	Pseudo LRM
PQ	Product Quality
PU	Production Unit
PUM	Product User Manual
QUID	QUality Information Document
RMS	Root Mean Square
RPC	Remote Protocol Client
RT	Real Time
S1A	Sentinel-1A
S1B	Sentinel-1B
S3A	Sentinel-3A
S3B	Sentinel-3B
S6A	Sentinel-6A
SAR	Synthetic Aperture Radar
SARIn	SAR Interferometric
SPC	SPeCtral
SRD	System Requirement Document
SWH	Significant Wave Height

SWON	SWOT nadir
TAC	Thematic Assembly Center
TDS	Thredds Data Server
VAVH	Visual AVerage Height
WMS	Web Map Service

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If needed, you can contact us .

1) INTRODUCTION

a) Summary

This document is one of the user manuals for the Copernicus Marine Service Near-Real-Time satellite wave products produced by the WAVE-TAC (Thematic Assembly Centre).

The WAVE TAC (Thematic Assembly Centre) is one of the eight TAC of the Copernicus Marine Service. The data produced in the frame of this TAC are generated by different processing systems including data from altimeter and other wave measuring devices (SAR, SWIM) for all the Copernicus missions (Sentinel-6A, Sentinel-3A/B, Sentinel-1A/B) and other collaborative or opportunity missions (e.g. : Jason-3, HaiYang-2 missions, Saral/AltiKa, Cryosat-2, CFOSAT). Through the WAVE-TAC, the Copernicus Marine Service also disseminates L3 products/datasets produced under Eumetsat responsibility. They mainly concern the Copernicus altimeter missions (Sentinel-3 & Sentinel-6).

This Product User Manual (PUM) details the following Level-3 product which is delivered in Near-Real-Time by the WAVE-TAC:

- WAVE_GLO_WAV_L3_SPC-FWK_NRT_014_002: Directional SPeCtral (SPC) integral parameter level-3 product based on observations of wave systems and their propagation over the ocean ('fireworks' algorithm). The observations are both SAR measurements on-board Sentinel-1A and Sentinel-1B missions and rotating Radar SWIM off-nadir measurements on-board CFOSAT mission.

2) DESCRIPTION OF THE PRODUCT SPECIFICATION

a) General Information

Product Lines	WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002
Geographical coverage	Global Ocean
Variables	The following integral parameters are used to describe observed swell fields along propagation time, from the storm source to the land: **VAVH**: sea surface significant wave height, i.e. the mean wave height of the highest third of the waves ($H_{1/3}$) for the considered wave partition **VTPK**: sea surface wave period at partition variance spectral density maximum, i.e. wave period at partition spectral peak / peak period (Tp) **VPED**: sea surface wave from direction at partition spectral peak, i.e. wave principal direction at partition spectral peak / peak direction (Dp) Where invalid, these variables are replaced by their fill values. Additional information at observation time also available in a separate group untitled "**obs_params**". Additional information for more expert users also available in a separate group untitled "**exp_params**". **Additional format:** This dataset is available in two different formats: • Files gathering only waves measurements belonging to the same swell field, covering vaious time periods of 10 to 15 days, • Files gathering only waves measurements belonging to all swell fields, but covering covering a 3 hour time period, Those two file formats contain the same information but arranged in two different ways in order to best satisfy all user needs.
Product Type	Near-Real-Time Level-3 observations and pseudo-observations (propagated over time)
Available time series	January 2020 – Present (depends on mission, see QUID)
Temporal resolution	3-hourly (one file contains one swell field propagated from one storm source until dissipation offshore or at the coast)
Update Frequency	Daily
Target delivery time	Daily: 12:00 UTC

Horizontal resolution	Discrete
Number of vertical levels	1 (Surface only)
Format	NetCDF-4

b) Production System Description

The SPeCtral FireWorKs system provides spectral integrated parameters of swell systems along their propagation path from the storm generation to the land. These parameters can be separated into observations and pseudo-observations. While the first ones are directly obtained from the wave spectrum partition of L2 satellite measurements, the pseudo-observations are derived using the 'Fireworks' algorithm. This algorithm propagates the observed partitions over the ocean using a linear wave propagation model from the observation time (both forwards and backwards). It then associates the propagated partitions by storm and filters out the partitions which do not pertain to any coherent swell system. The propagated partitions selected are then referred to as pseudo-observations.

WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002

Summary of the operational configuration:

Domain and resolution: The spectral observations and pseudo-observations are delivered over the global area. They are based on L2 Sentinel-1A and Sentinel-1B and L2P CFOSAT/SWIM wave measurements and available along each wave propagation path, from the generation storm to the land using a linear wave propagation model from observation time. The measurements are provided along the swell path over an irregular spatial grid with a regular time step (3h).

Geographic coverage: The geographic coverage is constrained by the availability of wave measurements propagating away from land and islands and in the absence of ice (updated daily from SEAICE_GLO_SEAICE_L4_NRT_OBSERVATIONS_011_001 product).
Only CFOSAT off-nadir measurements cover the European coasts, due to the absence of Sentinel-1 wave acquisitions in this area. European coasts are thus less well covered by this L3 product.

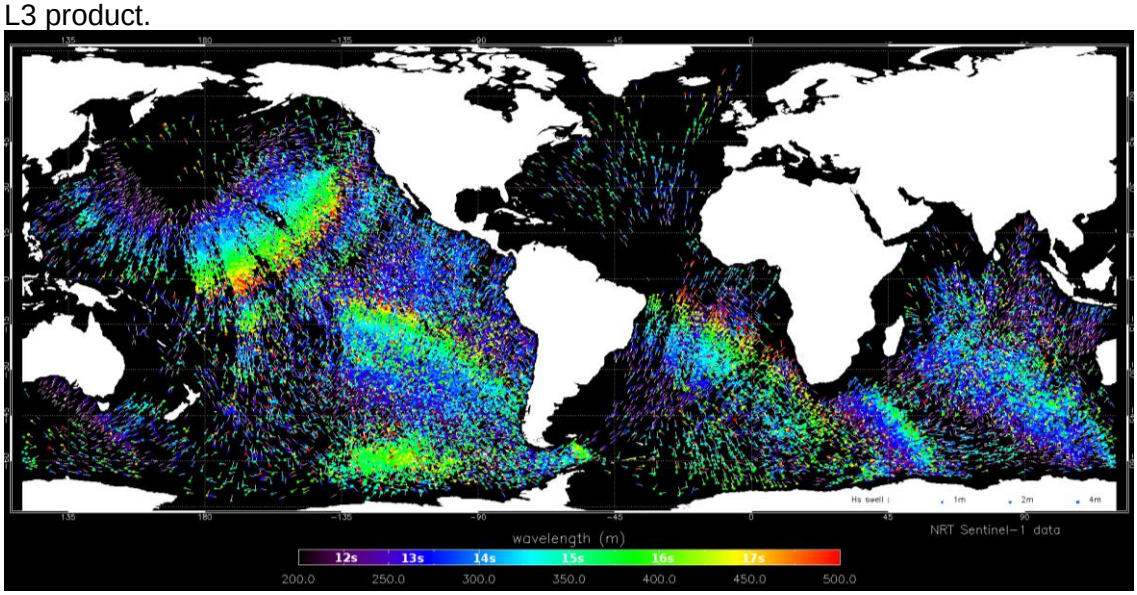

Figure 1: Example of wave measurements from Level-3 Sentinel-1A and Sentinel-1B SARs given on February 5th 2018.

c) Dataset details

Product Lines	WAVE_GLO_PHY_SPC-FWK_L3_NRT_014_002

Geographical coverage	Global
Datasets: cmems_obs-wave_glo_phy-spc-fwk_nrt_cfo-l3_PT15S-i cmems-obs-wave_glo_phy_spc-fwk_nrt_s1a_l3_PT15S-i cmems-obs-wave_glo_phy_spc-fwk_nrt_s1b_l3_PT15S-i	
Product Type	NRT Observations
Update frequency	Daily
Available time series	From 2020 up to real-time
Target delivery time	Daily at 12pm (noon) UTC
Temporal resolution	3 hourly instantaneous

Dataset : cmems_obs-wave_glo_phy-spc-fwk_nrt_cfo-l3_PT15S-i contains the observations and pseudo-observations over the swell propagation path based on CFOSAT measurements.
VAVH [m] Partition significant wave height along propagation sea_surface_wave_significant_height
VTPK [m] Partition peak period sea_surface_wave_period_at_variance_spectral_density_maximum
VPED [m] Partition peak direction along propagation (oceano. convention) sea_surface_wave_from_direction_at_variance_spectral_density_maximum
obs_params/VAVH_flag Flag for the sea_surface_swell_wave_part_significant_height
obs_params/VTPK_flag Flag for the sea_surface_swell_wave_part_peak_period
obs_params/VPED_flag Flag for the sea_surface_swell_wave_part_princ_from_direction
Dataset : cmems-obs-wave_glo_phy_spc-fwk_nrt_s1a_l3_PT15S-i contains the observations and pseudo-observations over the swell propagation path based on Sentinel-1 A measurements.
VAVH [m] Partition significant wave height along propagation sea_surface_wave_significant_height
VTPK [m] Partition peak period sea_surface_wave_period_at_variance_spectral_density_maximum
VPED [m] Partition peak direction along propagation (oceano. convention) sea_surface_wave_from_direction_at_variance_spectral_density_maximum
obs_params/VAVH_flag Flag for the sea_surface_swell_wave_part_significant_height
obs_params/VTPK_flag Flag for the sea_surface_swell_wave_part_peak_period
obs_params/VPED_flag Flag for the sea_surface_swell_wave_part_princ_from_direction
Dataset : cmems-obs-wave_glo_phy_spc-fwk_nrt_s1b_l3_PT15S-i

contains the observations and pseudo-observations over the swell propagation path based on Sentinel-1 B measurements.
VAVH [m] Partition significant wave height along propagation sea_surface_wave_significant_height
VTPK [m] Partition peak period sea_surface_wave_period_at_variance_spectral_density_maximum
VPED [m] Partition peak direction along propagation (oceano. convention) sea_surface_wave_from_direction_at_variance_spectral_density_maximum
obs_params/VAVH_flag Flag for the sea_surface_swell_wave_part_significant_height
obs_params/VTPK_flag Flag for the sea_surface_swell_wave_part_peak_period
obs_params/VPED_flag Flag for the sea_surface_swell_wave_part_princ_from_direction

d) Additional information on some parameters

All integral parameters are computed using a partitioned elevation spectrum for Sentinel-1. For CFOSAT, the partitioning algorithm cannot be applied on the elevation spectrum as it is too noisy. The partitioning algorithm is therefore applied on the slope spectrum (= elevation spectrum multiplied by the square of the wavenumber). Up to EIS November 2025, the integral parameters for CFOSAT were also computed using the slope spectrum, creating an artificial bias in wavelength comparisons between Sentinel-1 and CFOSAT observations. Starting from EIS November 2025, partitions for CFOSAT are still computed using the slope spectrum. However, the integral parameters of these partitions are now computed using the elevation spectrum, to ensure consistency with Sentinel-1 measurements.

e) Processing information

At each delivery, the swell field analysis is performed over the last 3 weeks. This is to take into account the possible swell propagation from the storm source for more than 10 days before reaching the coast plus a potential storm duration of several days.
After this swell field analysis, the older swell fields are almost unchanged from one update to the next as very few observations can belong to the oldest swell fields. For the more recent days, new swell fields can appear from one day to the next if enough L2 observations have been gathered to create a new swell system. Also, older swell fields can be enriched with additional recent L2 observations.

f) Catalogue updating

In Copernicus Marine catalogue, a purge is performed at each delivery in order to delete the oldest and least complete files for all L3 products. However, the user can also keep track of the successive updates, using the production date included in the file name.
The production chain runs once a day to produce level-3 SPC products from all latest available L2 measurements.
Level-3 files are updated by 12:00 UTC.

g) Time averaging

Instantaneous values are provided. There is no temporal averaging or interpolation performed.

h) Product use and impact

This product is related to the following Copernicus Marine Service products:

- WAVE_GLO_PHY_SPC_L3_NRT_014_009: L3 product containing spectral observations – both the integral parameters and the full 2D spectrum – based on CFOSAT/SWIM measurements (without propagation), This product basically gathers the CFOSAT/SWIM input for the ‘fireworks’ algorithm.
- WAVE_GLO_PHY_SPC-FWK_L4_NRT_014_004: L4 gridded product based on the L3 product described on this document.

3) FILE FORMAT

The products are stored using the NetCDF format.

To know more about the NetCDF format, please follow this link:
What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:
how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

4) FILES NOMENCLATURE

An exhaustive description of the files nomenclature when downloaded through the web portal is available here : Files Nomenclature and Format

More information on product and dataset nomenclatures is available here Products and Datasets nomenclature

The nomenclature of files of SPC-FWK_L3_NRT product is described below.

Storm associated swell field format:
dataset-wave-sar-l3-spc-nrt-global-<mission>_<datemin>_<datemax>_P<dateprod>_<storm_date>_lon_<storm_lon>_lat_<storm_lat>-<status>.nc

3-hour swell field format:
dataset-wave-sar-l3-spc-nrt-global--<mission>_<datemin>_<datemax>_P<dateprod>-3H-<status>.nc

With:

- <mission>= s1a, s1b or swi
- <datemin>= date of first measurement in the file YYYYmmddTHHMMZ
- <datemax>= date of last measurement in the file YYYYmmddTHHMMZ
- <dateprod>=date of production of the file YYYYmmddTHHMMZ
- <storm_date>=date of the generation storm source, dd-mmm-YYYY-hhH
- <storm_lon>=longitude of the storm source: E/WXXX [in degrees]
- <storm_lat>= latitude of the storm source: N/SXX [in degrees]
- <status> = "nrt" for the most recent swell fields that will be subject to updates as more swell observations can be gathered. "rep" for the oldest and most-up-to-date swell fields that gather all potential observations.

4.1) FILE SIZE

DATASET NAME	NAME OF FILE	DIMENSION [GB]
cmems-obs-wave_glo_phy_spc-fwk_nrt_cfo_l3_PT15S-i	dataset-wave-sar-l3-spc-nrt-global-swi_<datemin>_<datemax>_P<dateprod>_<storm_date>_lon_<storm_lon>_lat_<storm_lat>-<status>.nc	0.02
	dataset-wave-sar-l3-spc-nrt-global-swi_<datemin>_<datemax>_P<dateprod>-3H-<status>.nc	0.1
cmems-obs-wave_glo_phy_spc-fwk_nrt_s1a_l3_PT15S-i	dataset-wave-sar-l3-spc-nrt-global-s1a_<datemin>_<datemax>_P<dateprod>_<storm_date>_lon_<storm_lon>_lat_<storm_lat>-<status>.nc	0.02
	dataset-wave-sar-l3-spc-nrt-global-s1a_<datemin>_<datemax>_P<dateprod>-3H-<status>.nc	0.1
cmems-obs-wave_glo_phy_spc-fwk_nrt_s1b_l3_PT15S-i	dataset-wave-sar-l3-spc-nrt-global-s1b_<datemin>_<datemax>_P<dateprod>_<storm_date>_lon_<storm_lon>_lat_<storm_lat>-<status>.nc	0.02
	dataset-wave-sar-l3-spc-nrt-global-s1b_<datemin>_<datemax>_P<dateprod>-3H-<status>.nc	0.1

4.2) STRUCTURE OF FILES

Examples of the header of output NetCDF files are inserted in **annex**, for each dataset.

5) REFERENCES

Quality Information Document QUID: the associated QUID document for this product is CMEMS-WAVE-QUID-014-002.

6) ANNEX

The variables described in the netcdf files are organized in different ways depending on the level of information:

- The variables related to the propagated observations, these are given at the root of the netcdf file, as regular variables
- One group for the observed variables, called "obs_params", in which are gathered variables related to the L2 observations (such as the spectral information).
- One group of variables dedicated to more advanced users, called "exp_params", in which are gathered variables related to the storm event at the origin of the L2 observations (such as the storm location, times, dimensions...).
- The gobal variables, in which are gathered more general parameters such as the algorithm version, the data producer, the first and last time of the observations and propagated observations.

Example of instantaneous NetCDF metadata from these files is given below.
The NetCDF dumps for CFOSAT and S1 are slightly different. The two are provided below.

I S1 structure

```
netcdf dataset-wave-sar-l3-spc-nrt-global-s1a_20240508T1200Z_20240527T2100Z_P20240531T0916Z_09-MAY-2024-12H_lon_W167_lat_S58-nrt {
dimensions:
	obs = UNLIMITED ; // (1256 currently)
	spectrum_wavenumber = 60 ;
	spectrum_direction = 72 ;
	propag_time = 156 ;
	name_size = 67 ;
variables:
	double time(propag_time) ;
		time:units = "days since 1950-01-01T00:00:00Z" ;
		time:long_name = "Date of propagation in CNES convention" ;
		time:standard_name = "time" ;
	float longitude(propag_time, obs) ;
		longitude:_FillValue = -999.f ;
		longitude:units = "degrees_east" ;
		longitude:long_name = "Longitude of propagated observation" ;
		longitude:standard_name = "longitude" ;
		longitude:description = "East longitude relative to Greenwich meridian" ;
	float latitude(propag_time, obs) ;
		latitude:_FillValue = -999.f ;
		latitude:units = "degrees_north" ;
		latitude:long_name = "Latitude of propagated observation" ;
		latitude:standard_name = "latitude" ;
		latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
	int VAVH(propag_time, obs) ;
		VAVH:_FillValue = -999 ;
		VAVH:units = "m" ;
		VAVH:long_name = "Partition significant wave height along propagation" ;
		VAVH:standard_name = "sea_surface_wave_significant_height" ;
		VAVH:description = "The partition SWH is estimated along propagation before and after observations time, 3000 km from the storm source until it reaches land or propagates for too long. It uses decay and dissipation laws described in Ardhuin et al. 2009 and Collard et al. 2009" ;
		VAVH:scale_factor = 0.001 ;
	int VTPK(obs) ;
		VTPK:units = "s" ;
		VTPK:long_name = "Partition peak period" ;
		VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
```

```
                VTPK:description = "The partition peak period is assumed constant along the propagation (deep
water). It is equals to the partition peak period at observation time." ;
                VTPK:scale_factor = 0.01 ;
        int VPED(propag_time, obs) ;
                VPED:_FillValue = -999 ;
                VPED:units = "degree" ;
                VPED:long_name = "Partition peak direction along propagation (oceano. convention)" ;
                VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
                VPED:description = "The partition peak direction is estimated along propagation before and after
observations time, from the storm source until it reaches land or propagates for too long. It assumes great circle
propagation. It corresponds to the direction from which the waves arrive, in degrees from North, clockwise." ;
                VPED:scale_factor = 0.1 ;

// global attributes:
                :Processor_version = "-" ;
                :Comment = "Spectral wave parameters along propagation and at observation time (measured by
SAR) by swell field for satellite mission Sentinel-1A" ;
                :Conventions = "CF-1.6" ;
                :creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
                :references = "http://marine.copernicus.eu" ;
                :creator_url = "http://marine.copernicus.eu" ;
                :project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
                :contact = "servicedesk.cmems@mercator-ocean.eu" ;
                :processing_level = "L3" ;
                :creation_date = "20240531T0916Z" ;
                :keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
                :institution = "CLS" ;
                :creator_name = "CMEMS - Wave Thematic Assembly Center" ;
                :license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
                :standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard
Name Table v37" ;
                :title = "NRT Sentinel-1A Global Ocean Spectral Wave L3 product" ;
                :source_platform = "Sentinel-1A measurements" ;
                :based_on = "Sentinel-1A WV_L2_OCN" ;
                :first_obs_time = "20240508T164541Z" ;
                :last_obs_time = "20240515T162414Z" ;
                :first_propag_time = "20240508T1200Z" ;
                :last_propag_time = "20240527T2100Z" ;
                :Rough_storm_source_point_longitude = "-191.137 deg" ;
                :Rough_storm_source_point_latitude = " -55.705 deg" ;
                :Rough_storm_source_size = "1477.930 km" ;
                :Swell_generation_duration = "1.25 days" ;

group: obs_params {
  variables:
        double VAVH(obs) ;
                VAVH:units = "m" ;
                VAVH:long_name = "Partition significant wave height at observation time" ;
                VAVH:standard_name = "sea_surface_wave_significant_height" ;
                VAVH:description = "The partition SWH is retrieved from the L2 OCN WV products." ;
        float VPED(obs) ;
                VPED:units = "degree" ;
                VPED:long_name = "Partition peak direction at observation time (oceano. convention)" ;
                VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
                VPED:description = "The partition peak direction is retrieved from the L2 OCN WV products. It
corresponds to the direction from which the waves arrive, in degrees from North, clockwise." ;
        float VTPK(obs) ;
                VTPK:units = "s" ;
                VTPK:long_name = "Partition peak period" ;
                VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
                VTPK:description = "The partition peak period is retrieved from the L2 OCN WV products." ;
        double time(obs) ;
                time:units = "days since 1950-01-01T00:00:00Z" ;
                time:long_name = "Observation time" ;
                time:standard_name = "obs_time" ;
```

```
        float longitude(obs) ;
                longitude:units = "degrees_east" ;
                longitude:long_name = "Longitude of observation" ;
                longitude:standard_name = "obs_longitude" ;
                longitude:description = "East longitude relative to Greenwich meridian" ;
        float latitude(obs) ;
                latitude:units = "degrees_north" ;
                latitude:long_name = "Latitude of observation" ;
                latitude:standard_name = "obs_latitude" ;
                latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
        float platform_azimuth_angle(obs) ;
                platform_azimuth_angle:units = "degrees" ;
                platform_azimuth_angle:long_name = "Heading of observation" ;
                platform_azimuth_angle:standard_name = "platform_azimuth_angle" ;
                platform_azimuth_angle:description = "platform heading (azimuth from north)" ;
        float wave_spec(spectrum_wavenumber, spectrum_direction, obs) ;
                wave_spec:units = "m^4" ;
                wave_spec:long_name = "Estimated directional wave height spectrum" ;
                wave_spec:standard_name = "wave_spectrum" ;
        int partition_domain_spec(spectrum_wavenumber, spectrum_direction, obs) ;
                partition_domain_spec:units = "1" ;
                partition_domain_spec:long_name = "Mask of the partitions on wave spectrum" ;
                partition_domain_spec:standard_name = "wave_partition" ;
        float wavenumber_spec(spectrum_wavenumber) ;
                wavenumber_spec:units = "rad/m" ;
                wavenumber_spec:long_name = "Array of the wavenumbers for polar spectra" ;
                wavenumber_spec:standard_name = "wavenumber_spec" ;
        float direction_spec(spectrum_direction) ;
                direction_spec:units = "degree" ;
                direction_spec:long_name = "Array of angular values for polar spectra (degrees from North,
clockwise)" ;
                direction_spec:standard_name = "direction_spec" ;
        float nv(obs) ;
                nv:units = "1" ;
                nv:long_name = "Normalised variance estimated in the input L2 SLC image" ;
        float inten(obs) ;
                inten:units = "1" ;
                inten:long_name = "Input SLC image intensity estimated in the input L2 SLC image" ;
        float snr(obs) ;
                snr:units = "dB" ;
                snr:long_name = "Signal to noise ratio estimated in the input L2 cross-spectrum" ;
        float resolution_spec(spectrum_direction, obs) ;
                resolution_spec:units = "m" ;
                resolution_spec:long_name = "Azimuth cut-off wavelength as function of wave direction, for the input
L2 spectrum" ;
                resolution_spec:standard_name = "resolution_spec" ;
        float incidence_angle(obs) ;
                incidence_angle:_FillValue = -999.f ;
                incidence_angle:units = "degree" ;
                incidence_angle:long_name = "Observation incidence angle" ;
                incidence_angle:standard_name = "incidence_angle" ;
        int VPED_flag(obs) ;
                VPED_flag:units = "1" ;
                VPED_flag:long_name = "Global quality flag for each L2 swell partition (partition peak direction
coherency) - Values from 0 to 5" ;
                VPED_flag:standard_name =
"sea_surface_wave_from_direction_at_variance_spectral_density_maximum_flag" ;
                VPED_flag:description = "The quality of each L2 swell partition peak direction is examined through 5
successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
        int VTPK_flag(obs) ;
                VTPK_flag:units = "1" ;
                VTPK_flag:long_name = "Global quality flag for each L2 swell partition (partition peak period
coherency) - Values from 0 to 5" ;
                VTPK_flag:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_flag" ;
```

```
                VTPK_flag:description = "The quality of each L2 swell partition peak period is examined through 5
successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
        int VAVH_flag(obs) ;
                VAVH_flag:units = "1" ;
                VAVH_flag:long_name = "Global Hs quality flag for each L2 swell partition (partition Hs coherency) -
Values from 0 to 1" ;
                VAVH_flag:standard_name = "sea_surface_wave_significant_height_flag" ;
                VAVH_flag:description = "The quality of each L2 swell partition Hs is examined through successive
filters. Data quality goes from 0 (poor) to 1 (passed all the filters)." ;
        char L2_source_product(obs, name_size) ;
                L2_source_product:long_name = "L2 ncdf source product filename" ;
                L2_source_product:standard_name = "L2_source_product" ;
        byte L2_partition_quality_flag(obs) ;
                L2_partition_quality_flag:_FillValue = -128b ;
                L2_partition_quality_flag:long_name = "Quality flag per partition for each L2 swell partition" ;
                L2_partition_quality_flag:valid_range = 0b, 4b ;
                L2_partition_quality_flag:flag_values = 0b, 1b, 2b, 3b, 4b ;
                L2_partition_quality_flag:flag_meanings = "very_good good medium low poor" ;
 } // group obs_params

group: exp_params {
  variables:
        float foc_longitude(obs) ;
                foc_longitude:units = "degrees_east" ;
                foc_longitude:long_name = "Focal longitude" ;
                foc_longitude:standard_name = "foc_longitude" ;
                foc_longitude:description = "The focal longitude corresponds to the center longitude of all the
propagated observations that are refocusing at the same time step. It can differ among the set of propagated
observations." ;
        float foc_latitude(obs) ;
                foc_latitude:units = "degrees_north" ;
                foc_latitude:long_name = "Focal latitude" ;
                foc_latitude:standard_name = "foc_latitude" ;
                foc_latitude:description = "The focal latitude corresponds to the center latitude of all the propagated
observations that are refocusing at the same time step. It can differ among the set of propagated observations." ;
        float foc_size(obs) ;
                foc_size:units = "km" ;
                foc_size:long_name = "Focal size diameter" ;
                foc_size:standard_name = "foc_size" ;
                foc_size:description = "The focal size diameter corresponds to the diameter of the refocusing region
where all the propagated observations that are refocusing at the same time step have converged. It can differ among the
set of propagated observations." ;
        float foc_size_along(obs) ;
                foc_size_along:units = "km" ;
                foc_size_along:long_name = "Focal size diameter projected in main propagation direction" ;
                foc_size_along:standard_name = "foc_size_along" ;
                foc_size_along:description = "Idem but projected in main propagation direction. It can differ among
the set of propagated observations." ;
        float foc_size_across(obs) ;
                foc_size_across:units = "km" ;
                foc_size_across:long_name = "Focal size diameter projected in main transverse propagation
direction" ;
                foc_size_across:standard_name = "foc_size_across" ;
                foc_size_across:description = "Idem but projected in main transverse direction. It can differ among
the set of propagated observations." ;
        double foc_time(obs) ;
                foc_time:units = "days since 1950-01-01T00:00:00Z" ;
                foc_time:long_name = "Refocusing time in CNES convention" ;
                foc_time:standard_name = "foc_time" ;
                foc_time:description = "The refocusing time corresponds to the time when the propagated
observations are converging. It can differ among the set of propagated observations." ;
 } // group exp_params
}
```

II CFOSAT structure

```
netcdf dataset-wave-sar-l3-spc-nrt-global-swi__20240508T1200Z_20240527T2100Z_P20240517T0816Z_09-MAY-2024-12H_lon_W167_lat_S58-nrt {
dimensions:
obs = UNLIMITED ; // (2200 currently)
spectrum_wavenumber = 32 ;
spectrum_direction = 24 ;
propag_time = 156 ;
name_size = 56 ;
variables:
double time(propag_time) ;
time:units = "days since 1950-01-01T00:00:00Z" ;
time:long_name = "Date of propagation in CNES convention" ;
time:standard_name = "time" ;
float longitude(propag_time, obs) ;
longitude:_FillValue = -999.f ;
longitude:units = "degrees_east" ;
longitude:long_name = "Longitude of propagated observation" ;
longitude:standard_name = "longitude" ;
longitude:description = "East longitude relative to Greenwich meridian" ;
float latitude(propag_time, obs) ;
latitude:_FillValue = -999.f ;
latitude:units = "degrees_north" ;
latitude:long_name = "Latitude of propagated observation" ;
latitude:standard_name = "latitude" ;
latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
int VAVH(propag_time, obs) ;
VAVH:_FillValue = -999 ;
VAVH:units = "m" ;
VAVH:long_name = "Partition significant wave height along propagation" ;
VAVH:standard_name = "sea_surface_wave_significant_height" ;
VAVH:description = "The partition SWH is estimated along propagation before and after observations time, 3000 km from the storm source until it reaches land or propagates for too long. It uses decay and dissipation laws described in Ardhuin et al. 2009 and Collard et al. 2009" ;
VAVH:scale_factor = 0.001 ;
int VTPK(obs) ;
VTPK:units = "s" ;
VTPK:long_name = "Partition peak period" ;
VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
VTPK:description = "The partition peak period is assumed constant along the propagation (deep water). It is equals to the partition peak period at observation time." ;
VTPK:scale_factor = 0.01 ;
int VPED(propag_time, obs) ;
VPED:_FillValue = -999 ;
VPED:units = "degree" ;
VPED:long_name = "Partition peak direction along propagation (oceano. convention)" ;
VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
VPED:description = "The partition peak direction is estimated along propagation before and after observations time, from the storm source until it reaches land or propagates for too long. It assumes great circle propagation. It corresponds to the direction from which the waves arrive, in degrees from North, clockwise." ;
VPED:scale_factor = 0.1 ;

// global attributes:
:Processor_version = "-" ;
:Comment = "Spectral wave parameters along propagation and at observation time (measured by SAR) by swell field for satellite mission Swim" ;
:Conventions = "CF-1.6" ;
:creator_email = "servicedesk.cmems@mercator-ocean.eu" ;
```

```
:references = "http://marine.copernicus.eu" ;
:creator_url = "http://marine.copernicus.eu" ;
:project = "COPERNICUS MARINE ENVIRONMENT MONITORING SERVICE (CMEMS)" ;
:contact = "servicedesk.cmems@mercator-ocean.eu" ;
:processing_level = "L3" ;
:creation_date = "20240517T0816Z" ;
:keywords_vocabulary = "NetCDF COARDS Climate and Forecast Standard Names" ;
:institution = "CLS" ;
:creator_name = "CMEMS - Wave Thematic Assembly Center" ;
:license = "http://marine.copernicus.eu/web/27-service-commitments-and-licence.php" ;
:standard_name_vocabulary = "NetCDF Climate and Forecast (CF) Metadata Convention Standard Name Table v37" ;
:title = "NRT Swim Global Ocean Spectral Wave L3 product" ;
:source_platform = "Swim measurements" ;
:based_on = "Swim SW_L2_OCN" ;
:first_obs_time = "20240508T172234Z" ;
:last_obs_time = "20240515T170722Z" ;
:first_propag_time = "20240508T1200Z" ;
:last_propag_time = "20240527T2100Z" ;
:Rough_storm_source_point_longitude = "-191.137 deg" ;
:Rough_storm_source_point_latitude = " -55.705 deg" ;
:Rough_storm_source_size = "1477.930 km" ;
:Swell_generation_duration = "1.25 days" ;

group: obs_params {
  variables:
        double VAVH(obs) ;
                VAVH:units = "m" ;
                VAVH:long_name = "Partition significant wave height at observation time" ;
                VAVH:standard_name = "sea_surface_wave_significant_height" ;
                VAVH:description = "The partition SWH is retrieved from the L2 OCN WV products." ;
        float VPED(obs) ;
                VPED:units = "degree" ;
                VPED:long_name = "Partition peak direction at observation time (oceano. convention)" ;
                VPED:standard_name = "sea_surface_wave_from_direction_at_variance_spectral_density_maximum" ;
                VPED:description = "The partition peak direction is retrieved from the L2 OCN WV products. It
corresponds to the direction from which the waves arrive, in degrees from North, clockwise." ;
        float VTPK(obs) ;
                VTPK:units = "s" ;
                VTPK:long_name = "Partition peak period" ;
                VTPK:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum" ;
                VTPK:description = "The partition peak period is retrieved from the L2 OCN WV products." ;
        double time(obs) ;
                time:units = "days since 1950-01-01T00:00:00Z" ;
                time:long_name = "Observation time" ;
                time:standard_name = "obs_time" ;
        float longitude(obs) ;
                longitude:units = "degrees_east" ;
                longitude:long_name = "Longitude of observation" ;
                longitude:standard_name = "obs_longitude" ;
                longitude:description = "East longitude relative to Greenwich meridian" ;
        float latitude(obs) ;
                latitude:units = "degrees_north" ;
                latitude:long_name = "Latitude of observation" ;
                latitude:standard_name = "obs_latitude" ;
                latitude:description = "Positive latitude is North latitude, negative latitude is South latitude." ;
        float platform_azimuth_angle(obs) ;
                platform_azimuth_angle:units = "degrees" ;
                platform_azimuth_angle:long_name = "Heading of observation" ;
                platform_azimuth_angle:standard_name = "platform_azimuth_angle" ;
                platform_azimuth_angle:description = "platform heading (azimuth from north)" ;
```

```
        float wave_spec(spectrum_wavenumber, spectrum_direction, obs) ;
                wave_spec:units = "m^2/rad" ;
                wave_spec:long_name = "Estimated directional wave slope spectrum for spectral beam 10 degrees" ;
                wave_spec:standard_name = "wave_spectrum" ;
        int partition_domain_spec(spectrum_wavenumber, spectrum_direction, obs) ;
                partition_domain_spec:units = "1" ;
                partition_domain_spec:long_name = "Mask of the partitions on wave spectrum" ;
                partition_domain_spec:standard_name = "wave_partition" ;
        float wavenumber_spec(spectrum_wavenumber) ;
                wavenumber_spec:units = "rad/m" ;
                wavenumber_spec:long_name = "Array of the wavenumbers for polar spectra" ;
                wavenumber_spec:standard_name = "wavenumber_spec" ;
        float direction_spec(spectrum_direction) ;
                direction_spec:units = "degree" ;
                direction_spec:long_name = "Array of angular values for polar spectra (degrees from North,
clockwise)" ;
                direction_spec:standard_name = "direction_spec" ;
        int VPED_flag(obs) ;
                VPED_flag:units = "1" ;
                VPED_flag:long_name = "Global quality flag for each L2 swell partition (partition peak direction
coherency) - Values from 0 to 5" ;
                VPED_flag:standard_name =
"sea_surface_wave_from_direction_at_variance_spectral_density_maximum_flag" ;
                VPED_flag:description = "The quality of each L2 swell partition peak direction is examined through 5
successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
        int VTPK_flag(obs) ;
                VTPK_flag:units = "1" ;
                VTPK_flag:long_name = "Global quality flag for each L2 swell partition (partition peak period
coherency) - Values from 0 to 5" ;
                VTPK_flag:standard_name = "sea_surface_wave_period_at_variance_spectral_density_maximum_flag" ;
                VTPK_flag:description = "The quality of each L2 swell partition peak period is examined through 5
successive filters. Data quality goes from 0 (poor) to 5 (passed all the filters)." ;
        int VAVH_flag(obs) ;
                VAVH_flag:units = "1" ;
                VAVH_flag:long_name = "Global Hs quality flag for each L2 swell partition (partition Hs coherency) -
Values from 0 to 1" ;
                VAVH_flag:standard_name = "sea_surface_wave_significant_height_flag" ;
                VAVH_flag:description = "The quality of each L2 swell partition Hs is examined through successive
filters. Data quality goes from 0 (poor) to 1 (passed all the filters)." ;
        int n_posneg(obs) ;
                n_posneg:_FillValue = -1 ;
                n_posneg:units = "1" ;
                n_posneg:long_name = "L2 ncdf posneg (only for SWIM source)" ;
                n_posneg:standard_name = "n_posneg" ;
        int n_box(obs) ;
                n_box:_FillValue = -1 ;
                n_box:units = "1" ;
                n_box:long_name = "L2 ncdf box (only for SWIM source)" ;
                n_box:standard_name = "n_posneg" ;
        char L2_source_product(obs, name_size) ;
                L2_source_product:long_name = "L2 ncdf source product filename" ;
                L2_source_product:standard_name = "L2_source_product" ;
 } // group obs_params

group: exp_params {
 variables:
        float foc_longitude(obs) ;
                foc_longitude:units = "degrees_east" ;
                foc_longitude:long_name = "Focal longitude" ;
                foc_longitude:standard_name = "foc_longitude" ;
```

```
                foc_longitude:description = "The focal longitude corresponds to the center longitude of all the
propagated observations that are refocusing at the same time step. It can differ among the set of propagated
observations." ;
        float foc_latitude(obs) ;
                foc_latitude:units = "degrees_north" ;
                foc_latitude:long_name = "Focal latitude" ;
                foc_latitude:standard_name = "foc_latitude" ;
                foc_latitude:description = "The focal latitude corresponds to the center latitude of all the propagated
observations that are refocusing at the same time step. It can differ among the set of propagated observations." ;
        float foc_size(obs) ;
                foc_size:units = "km" ;
                foc_size:long_name = "Focal size diameter" ;
                foc_size:standard_name = "foc_size" ;
                foc_size:description = "The focal size diameter corresponds to the diameter of the refocusing region
where all the propagated observations that are refocusing at the same time step have converged. It can differ among the
set of propagated observations." ;
        float foc_size_along(obs) ;
                foc_size_along:units = "km" ;
                foc_size_along:long_name = "Focal size diameter projected in main propagation direction" ;
                foc_size_along:standard_name = "foc_size_along" ;
                foc_size_along:description = "Idem but projected in main propagation direction. It can differ among
the set of propagated observations." ;
        float foc_size_across(obs) ;
                foc_size_across:units = "km" ;
                foc_size_across:long_name = "Focal size diameter projected in main transverse propagation
direction" ;
                foc_size_across:standard_name = "foc_size_across" ;
                foc_size_across:description = "Idem but projected in main transverse direction. It can differ among
the set of propagated observations." ;
        double foc_time(obs) ;
                foc_time:units = "days since 1950-01-01T00:00:00Z" ;
                foc_time:long_name = "Refocusing time in CNES convention" ;
                foc_time:standard_name = "foc_time" ;
                foc_time:description = "The refocusing time corresponds to the time when the propagated
observations are converging. It can differ among the set of propagated observations." ;
 } // group exp_params
```